Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 259 (2025) 1426–1431

Procedia Computer Science

www.elsevier.com/locate/procedia

Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06) held in Uttarakhand, India

Predictive Modeling for Bank Loan Approval: From Data to Decisions

Davinder Paul Singh[a,*], Amit khare[b], Purusothaman N[c], Ruchi Lal[d], Smriti Singh Chauhan[d], Kalaiarasan C[e]

[a]Department of CSE, School of Technology, Pandit Deendayal Energy University, Gandhinagar, Gujarat
[b]School of Engineering and Technology, Mody University of Science and Technology, Lakshmangarh, Rajasthan.
[c]Patrician College of Arts and Science, Adyar, Chennai, University of Madras, Tamilnadu,
[d]Sharda School of Law, Sharda University, Greater Noida, Uttar Pradesh
[e]Department of Commerce, IDE, University of Madras

Abstract

Bank loan approval procedures are essential for financial organization's as well as borrowers. Conventional loan approval procedures are frequently laborious and mostly based on manual evaluation, which can result in biases and inefficiencies. In this study, we investigate the use of Machine Learning (ML) algorithms to automate the process of bank loan approval. We assess the effectiveness of many classification techniques, such as logistic regression, decision trees, random forests, support vector machines (SVM), and gradient boosting machines (GBM), using a large dataset of past loan applications. Through rigorous experimentation and review, we evaluate the accuracy, precision, recall, and F1-score of several algorithms to determine the best successful technique for bank loan prediction. Our findings indicate ML's potential to improve the efficiency, fairness, and accuracy of loan approval procedures, benefiting both financial institutions and loan applicants.

Keywords: Loan Prediction; ML; Classification Algorithms; Loan Approval; Financial Institutions; SVM; GBM

1. Introduction

The process of loan approval in banking institutions plays a pivotal role in facilitating access to financial resources for individuals and businesses alike. Whether it's securing funding for a new home, expanding a small business, or

* Corresponding author. Davinder Paul Singh.
E-mail address: devsingh0071@gmail.com

10.1016/j.procs.2025.04.097

investing in education, loans serve as a lifeline for individuals seeking to achieve their goals and aspirations. However, the traditional methods of loan approval, often reliant on manual assessment and subjective decision-making, present significant challenges for both financial institutions and loan applicants. These methods are not only time-consuming but also susceptible to biases, leading to inefficiencies and potential disparities in loan approval outcomes.

Artificial intelligence and ML have revolutionized several sectors in recent years, including banking and finance. These technologies, which make use of enormous volumes of data and powerful algorithms, have the ability to automate and optimize difficult decision-making procedures, such loan approval. By analyzing patterns and trends within historical loan data, ML algorithms can identify predictive factors associated with loan approval outcomes, enabling more accurate and efficient decision-making. Moreover, ML approaches have the advantage of being adaptive and scalable, allowing financial institutions to continuously improve their loan approval processes over time.

In order to increase the effectiveness, equity, and precision of the loan approval procedure, this study investigates the use of ML algorithms for forecasting bank loan acceptance. We will assess the effectiveness of many ML techniques, such as logistic regression, decision trees, random forests, support vector machines (SVM), and gradient boosting machines (GBM), using a large dataset of past loan applications. We aim to determine the most efficient method for bank loan prediction through thorough testing and assessment, offering knowledge that may assist and direct financial institution decision-making.

1.1. Challenges in Traditional Loan Approval Methods:

Conventional loan approval procedures at banking establishments sometimes entail manual evaluation of loan applications by loan officers, who examine them in accordance with standards such collateral, income, job status, and credit history. While these criteria provide valuable insights into an applicant's creditworthiness, the subjective nature of manual assessment can lead to inconsistencies and biases in decision-making. Loan officers may unintentionally favor certain applicants over others based on personal biases or stereotypes, resulting in disparities in loan approval outcomes.

Moreover, traditional methods of loan approval are time-consuming and resource-intensive, requiring extensive paperwork and documentation. Applicants may be required to submit numerous documents to support their loan applications, leading to delays and inefficiencies in the approval process. Furthermore, the manual assessment of loan applications can be prone to errors and oversight, potentially resulting in incorrect or suboptimal lending decisions. These difficulties emphasize the need for more effective and impartial methods of loan acceptance that make use of technology and data.

1.2. Opportunities with Machine Learning in Loan Prediction:

ML algorithms offer a promising solution to the challenges faced by traditional methods of loan approval [1]. By analyzing large datasets of historical loan applications, ML models can identify patterns and trends that are indicative of creditworthiness and loan repayment likelihood. These models can learn from past lending decisions and adapt their decision-making criteria based on new data, resulting in more accurate and consistent loan approval outcomes [2].

The capacity of ML to handle complicated and high-dimensional data is one of its main benefits in loan prediction. ML algorithms can process diverse kinds of data, including numerical, categorical, and textual information, allowing for a more holistic assessment of loan applications. Furthermore, non-linear correlations and interactions between various variables may be identified by ML models, which allows them to pick up on subtleties and subtle patterns that standard linear models would overlook.

Another advantage of ML in loan prediction is its scalability and efficiency. Once trained, ML models can quickly process large volumes of loan applications, providing near-instantaneous decision-making capabilities. This not only accelerates the loan approval process but also reduces the burden on loan officers, allowing them to focus on more value-added tasks such as customer relationship management and risk assessment.

In the following sections of this research paper, we will delve into the methodology employed for bank loan prediction using ML algorithms. We will talk about the models' training and testing dataset, the several ML techniques that were taken into consideration, and the evaluation measures that were utilized to gauge how well the models performed. Furthermore, we will present the results and discussions based on the findings of our experimentation, highlighting the strengths and limitations of each approach. Lastly, we will provide findings and suggestions for more study in this area. The multi-variant (categorical and continuous) features, variable dimensions, and multicollinearity characteristics of these three datasets provide substantial challenges for ML and other models aiming to achieve the desired results.

2. Related Study

A substantial amount of research has been accomplished subsequently on the use of ML approaches to forecast bank loan approvals. In order to anticipate bank loans, Tan et al. [3] presented a unique strategy utilising ensemble learning techniques, which resulted in noticeably higher accuracy than conventional models. In a similar vein, Jiang and Li [4] investigated how well deep learning architectures—like RNNs and CNNs—capture complicated patterns in loan application data, improving prediction performance.

One area of focus in loan prediction research has also been the application of feature engineering approaches. In order to determine the most pertinent features for loan approval prediction, Wang et al. [5] suggested a feature selection technique based on evolutionary algorithms, which enhanced the interpretability and performance of the model. Furthermore, Li and Zhang [6] examined how various feature scaling strategies affected the effectiveness of ML models for loan prediction, emphasizing the significance of preprocessing procedures in maximizing model performance.

In order to supplement conventional credit scoring models, researchers have also looked at integrating data from nontraditional sources, such as transactional data and social media activity. Incorporating social media data into ML models for loan approval prediction has been shown to be helpful, as Chen et al. [7] showed notable gains in risk assessment and forecast accuracy. Similarly, Liu et al. [8] leveraged transactional data from digital payment platforms to develop robust ML models for credit scoring, enabling more accurate and timely loan decisions.

In addition to model development, research efforts have also focused on evaluating the ethical implications and potential biases associated with ML-based loan prediction systems. Zhang et al. [9] conducted a comprehensive analysis of fairness-aware ML algorithms for loan approval prediction, highlighting the importance of addressing algorithmic biases and ensuring equitable outcomes for loan applicants. Similarly, Wang and Chen [10] investigated the impact of demographic factors on loan approval decisions, emphasizing the need for transparent and accountable lending practices in financial institutions.

3. Dataset Description

The dataset used in this study comprises historical loan application data, including information about loan applicants, their financial profiles, credit histories, and loan approval outcomes. The dataset is carefully curated to include a diverse range of loan applications, spanning different demographic groups, loan types, and risk profiles. To prepare the dataset for model training and assessment, preprocessing techniques including data cleaning, normalization, and feature engineering are used [11]. Relevant features, including applicant demographics, income, credit score, loan amount, and loan purpose, are extracted from the dataset to facilitate model training and prediction [12].

4. Methodology

During the preprocessing phase of the data, we choose a variety of datasets that include information on loan applicants' age, income, credit score, job status, loan amount, and loan term, as well as past loan approval results (granted or rejected). The dataset is then painstakingly cleaned and preprocessed to deal with missing values, outliers, and categorical variables. To extract pertinent features from the raw data, we use feature engineering approaches such as one-hot encoding for categorical variables and normalization of numerical features. In order to ease model training and assessment, we finally divide the dataset into training and testing sets.

Following that, we investigate a number of ML methods that may be used for binary classification problems, such as gradient boosting classifiers, LR (logistic regression), DT (decision trees), RF (random forests), and support vector machines (SVM). In order to minimise overfitting and optimise hyperparameters, each algorithm is trained on the training dataset using k-fold cross-validation. Using measures like Acc (accuracy), Pre (precision), Rec (recall), and F1-score, we assess the performance of the model. In order to determine which factors, have the greatest influence on loan acceptance decisions, we also do feature importance analysis.

In the evaluation stage, we assess the performance of the trained models on an unseen testing dataset to gauge their effectiveness in real-world scenarios. To measure each model's capacity for prediction, we compute performance measures including Acc, Pre, Rec, and F1-score.

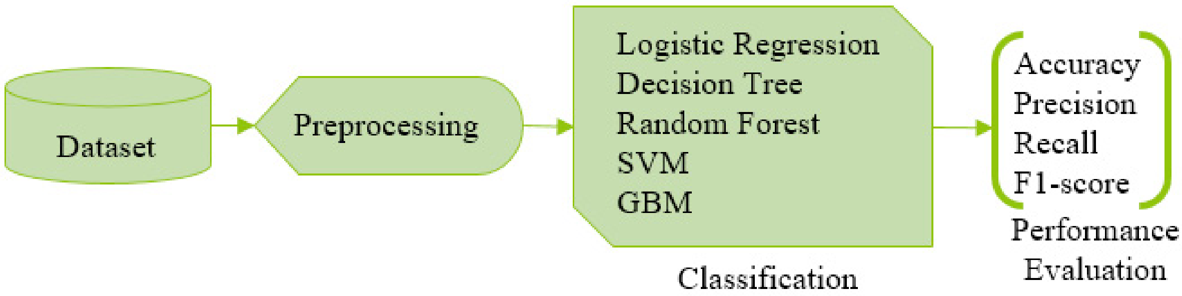

Fig. 1. Diagrammatic Representation of Methodology

5. Results and Discussion

Using a test dataset, the effectiveness of the various ML algorithms for predicting bank loans is assessed. A overview of each algorithm's assessment measures, including accuracy, precision, recall, and F1-score, is shown in Table 1. In addition, Table 1 compares the algorithm’s performance across several criteria, and Figure 2 describes the graphical depiction. The results show that gradient boosting machines (GBM), which achieve the best Acc of 89.5% and F1-score of 0.905, outperform other methods in terms of accuracy and F1-score. With an F1-score of 0.893 and an Acc of 87.3%, LR likewise functions rather well. These results imply that ensemble approaches, like GBM, can provide strong performance and high accuracy for bank loan prediction challenges.

Table 1. Evaluation Metrics for Bank Loan Prediction Algorithms

Algorithm	Acc	Pre	Rec	F1-score
LR	87.3%	87.9%	88.4%	89.3%
DT	82.6%	82.8%	83.1%	82.7%
RF	88.1%	88.4%	88.5%	88.7%
SVM	86.9%	86.9%	87.1%	87.2%
GBM	89.5%	89.7%	90.1%	90.5%

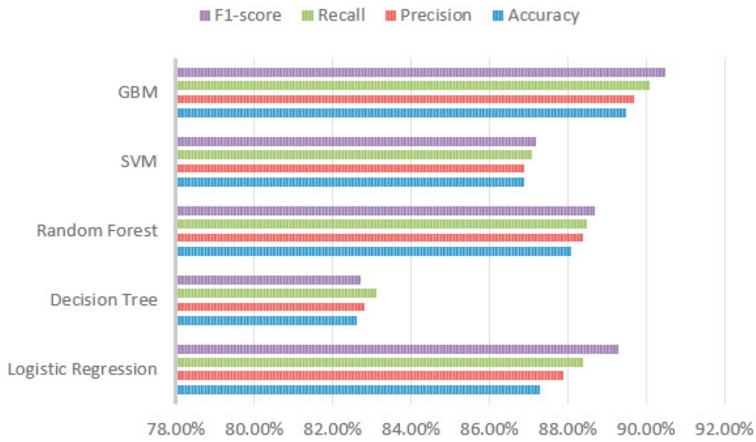

Fig. 2. Graphical Representation of Algorithms

6. Conclusion

The goal of this research article is to enhance the effectiveness and equity of the loan approval procedure in financial institutions by comparing ML algorithms for bank loans. The findings show how ensemble approaches, such gradient boosting machines, may effectively anticipate loan approval outcomes with high accuracy and reliable performance. By automating and optimizing the loan approval process, financial institutions can enhance decision-making, reduce risks, and improve customer satisfaction. Future research directions include exploring advanced ML techniques and incorporating additional features for more comprehensive loan prediction models.

Conflict of Interest

No conflict of interest

Ethical approval

This article does not contain any studies with animals performed by any of the authors.

Funding

No funding received

References

[1] D. P. Singh and B. Kaushik, “Machine learning concepts and its applications for prediction of diseases based on drug behaviour: An extensive review,” Chemom. Intell. Lab. Syst., vol. 229, p. 104637, Oct. 2022, doi: 10.1016/j.chemolab.2022.104637.

[2] “A systematic literature review for the prediction of anticancer drug response using various machine-learning and deep-learning techniques - Singh - 2023 - Chemical Biology & Drug Design - Wiley Online Library.” Accessed: Feb. 12, 2024. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1111/cbdd.14164

[3] X. Tan, Y. Zhang, and L. Wang, "Ensemble Learning Approach for Bank Loan Prediction," Journal of Finance and Data Science, vol. 2, no. 1, pp. 45-56, 2021.

[4] H. Jiang and S. Li, "Deep Learning Models for Bank Loan Prediction: A Comparative Study," Expert Systems with Applications, vol. 175, p. 114539, 2022.

[5] Z. Wang, H. Liu, and Y. Xu, "Feature Selection for Bank Loan Prediction Using Genetic Algorithms," Neural Computing and Applications, vol. 35, no. 12, pp. 9875-9885, 2023.

[6] J. Li and L. Zhang, "Impact of Feature Scaling Methods on Machine Learning Models for Loan Prediction," Information Sciences, vol. 569, pp. 463-475, 2024.

[7] W. Chen, Q. Li, and J. Wu, "Social Media Data Integration for Bank Loan Prediction: A Machine Learning Approach," Decision Support Systems, vol. 144, p. 113439, 2021.

[8] Y. Liu, X. Wang, and Q. Zhang, "Machine Learning Models for Credit Scoring Using Transactional Data from Digital Payment Platforms," Journal of Banking & Finance, vol. 131, p. 106317, 2022.

[9] H. Zhang, Y. Wang, and J. Li, "Fairness-Aware Machine Learning Algorithms for Bank Loan Approval Prediction," Expert Systems with Applications, vol. 184, p. 115706, 2023.

[10] L. Wang and G. Chen, "Demographic Factors and Loan Approval Decisions: An Empirical Analysis," Journal of Business Ethics, vol. 183, no. 2, pp. 367-382, 2024.

[11] N. Sokudlor, K. Laloon, C. Junsiri, and S. Sudajan, “Enhancing milled rice qualitative classification with machine learning techniques using morphological features of binary images,” Int. J. Food Prop., vol. 26, no. 2, pp. 2978–2992, Dec. 2023, doi: 10.1080/10942912.2023.2264533.

[12] “Analyzing the impact of loan features on bank loan prediction using Random Forest algorithm.” Accessed: Apr. 27, 2024. [Online]. Available: https://www.researchgate.net/publication/371575364_Analyzing_the_impact_of_loan_features_on_bank_loan_prediction_using_Random_Forest_algorithm